

UNIVERSITI TUNKU ABDUL RAHMAN

ACADEMIC YEAR 2024/2025

MAY EXAMINATION

UCCD1213/UCCG1213 FUNDAMENTALS OF DIGITAL MEDIA TECHNOLOGY

THURSDAY, 29 MAY 2025

TIME : 9.00 AM – 11.00 AM (2 HOURS)

EXAMINATION FOR BACHELOR DEGREE

Instruction to Candidates :

This question paper consists of **THREE (3)** questions in **Section A** and **TWO (2)** questions in **Section B**.

Answer **ALL** questions in **Section A** and **ONLY ONE (1)** question in **Section B**.
Each question carries 25 marks.

Should a candidate answer more than **ONE (1)** question in **Section B**, marks will only be awarded for the **FIRST** question in that section in the order the candidate submits the answers.

Candidates are allowed to use calculator.

Answer questions only in the answer booklet provided.

UCCD1213/UCCG1213 FUNDAMENTALS OF DIGITAL MEDIA TECHNOLOGY**Section A [Total: 75 marks]**

This section consists of **THREE (3) COMPULSORY** questions.

Q1. (a) Multimedia has permeated every part of our society. As the 21st century progresses further, it is more important that users understand multimedia and transition from passive consumers to active, thinking users. Briefly explain the differences between: -

- (i) hypertext and hypermedia. (4 marks)
- (ii) animation and video. (4 marks)
- (iii) interactive and non-interactive multimedia. (4 marks)
- (iv) captured media and synthesis media. (4 marks)

(b)

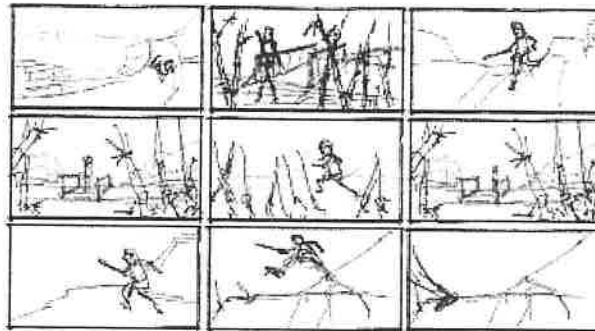


Figure 1.0: An Example of Storyboard

The first key to a successful multimedia project is to know where you are starting from, and what the project should accomplish when it is completed. One of the deliverables during the planning phase is storyboard, as depicted in Figure 1.0. Briefly describe **ANY THREE (3)** benefits of producing storyboard.

(9 marks)

[Total : 25 marks]

Q2. (a)



Figure 2.0: Multimedia Presentation

A multimedia presentation (Figure 2.0) is a dynamic way to share information, stories, or ideas. It combines various media forms, such as text, images, sound, and video, to create a more engaging and informative experience than traditional, single-medium presentations. One application area where multimedia presentations are increasingly and popularly used is in the education sector.

- (i) Describe **ANY THREE (3)** advantages of multimedia presentation in the education sector. (9 marks)
 - (ii) Explain **ANY ONE (1)** disadvantage of multimedia presentation in education sector. (3 marks)
- (b) The distinction between authoring tools and programming tools is not clear-cut. Generally, authoring tools are classified as ‘high level tools’ whereas programming languages are known as ‘low level tools’. Besides this classification, briefly elaborate **ANY TWO (2)** differences between ‘authoring tools’ and ‘programming languages’. (4 marks)
- (c) A project requires a plan and consideration of the necessary processes to carry out the plan; which constitutes the development cycle. Multimedia projects are by nature interdisciplinary and usually involve a team of people. Illustrate **ANY THREE (3)** reasons why good cooperation and coordination among team members are crucial to the success of a multimedia project. (9 marks)
- [Total : 25 marks]

UCCD1213/UCCG1213 FUNDAMENTALS OF DIGITAL MEDIA TECHNOLOGY

- Q3. (a) Computer and devices use two methods to represent fonts, namely bitmapped font and vector graphic font. Elaborate **ANY THREE (3)** key advantages of vector graphic font against bitmapped font. (9 marks)
- (b) When designing text for any multimedia presentation, one of the factors to consider is to strike a density balance in the design. Discuss **ANY THREE (3)** reasons why it is important to consider this factor. (9 marks)
- (c) The computer screen below (Figure 3.0) shows an animation of a cyclist cycling repeatedly from point A to point B and return to point A.

**Point B****Point A**

Figure 3.0: Animation of a Cyclist

While the cyclist is cycling and producing some soft cycling noise, clicking the body of the cyclist stops the cycling motion. Suggest the file format for the animation above. Justify your answers. (7 marks)

[Total : 25 marks]

Section B [Total: 25 marks]

This section consists of **TWO (2) OPTIONAL** questions. Answer **ANY ONE (1)** question.

Q4. (a)

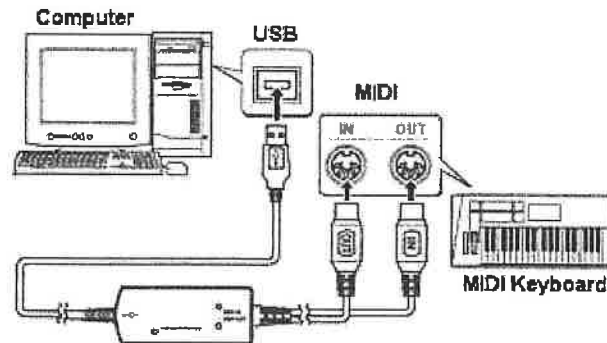


Figure 4.0: An Example of MIDI setup

MIDI, short for "Musical Instrument Digital Interface", is a protocol that allows electronic musical instruments, computers and other devices to communicate with one another. Elaborate **ANY FOUR (4)** fundamental differences between MIDI and digital audio. (8 marks)

- (b) A song that lasts 5 minutes is to be transmitted over a network. The music has been recorded at compact disk (CD) quality (44.1 kHz, 16 bit, Mono). Calculate the number of bytes required to store a full performance at CD quality audio. (5 marks)
NOTE: kHz – kilohertz, bit – binary digit
- (c) Briefly elaborate **ANY TWO (2)** reasons why it is important to identify the need for the use of diverse media elements during the user interface design in any multimedia-based applications. (6 marks)
- (d) Multimedia designer is responsible for the overall aesthetic and user experience of a multimedia project presentation. These include aspects on screen colours, visual consistency, meaningful icons, etc. Using appropriate examples, explain **ANY TWO (2)** reasons why *visual consistency* is important in the user interface design for any multimedia applications. (6 marks)

[Total : 25 marks]

UCCD1213/UCCG1213 FUNDAMENTALS OF DIGITAL MEDIA TECHNOLOGY

Q5. Virtual Reality (VR) is the creation of a virtual environment presented to our senses in such a way that we experience it as if we were really there. It uses a host of technologies to achieve goal and technically complex feat that has to account for our perception and cognition.

- (a) Explain **ANY TWO (2)** differences between *VR* and *Second Person VR*.
(6 marks)
- (b) What are the **THREE (3)** key challenges in implementing *Second Person VR*?
(9 marks)
- (c) Discuss **ANY TWO (2)** differences between VR and MR (Mixed Reality).
(6 marks)
- (d) What is the use of the device depicted in Figure 5.0.
(4 marks)



Figure 5.0 Head Mounted Display

[Total : 25 marks]